

MetroQ, says, "Apart from universal serial buses (USBs), manufacturers are coming up with Wi-Fi-enabled multimeters. A year or two down the line, connectivity in multimeters will become standard."

Rajesh Sawant, business development manager, Keysight Technologies India Pvt Ltd, too feels that connectivity is an important feature for multimeters. Talking about how it benefits users, he says, "Communication of various instruments on a common platform via wireless connectivity enables remote operations, monitoring and logging of parameters, thereby enhancing the productivity of the user."

A multimeter normally does not show a waveform. Therefore in some multimeters, an oscilloscope is in-built for waveform analysis, informs Kwatra. He says, "These scopemeters (multimeters + oscilloscopes) generally offer 20MHz to 25MHz bandwidth." Today's multimeters use advanced microcontrollers or application-specific integrated circuits (ASICs) to be able to support more functions accurately, notes Chandmal Goliya, director, KUSAM-MECO.

What is new in multimeters, feature-wise

Wireless connectivity and data logging. Some T&M companies have launched next-generation wireless connectivity solutions for digital multimeters that let you measure over a greater expanse, quicker and trouble-free. For instance, Fluke's CNX 3000 wireless multimeter allows you to access live measurements from three wireless modules, apart from the meter reading, from up to 20 metres away. This helps in seeing cause and effect in a chain of events, interactions between inputs/outputs or other measurement points, remotely as well as simultaneously.

In the case of Keysight's U1200 series digital multimeters, a user can add Bluetooth connectivity by attaching U1117A IR-to-Bluetooth adaptor

Point of view: Other trends observed in digital multimeters

Information (and not just data). Users of benchtop multimeters are now seeking solutions that offer them information rather than plain raw data. For instance, real-time statistical analysis on Rigol's DM3000 series offers statistical assessment on power factor, power ratio in dB or dBm, real-time histograms and trend displays. Engineers can now use this information to aid and improve their designs.

Multimeters going beyond electrical labs. Benchtop multimeters are no longer limited to measuring basic electric/electronic elements. Rigol's DM3068, for example, offers any-sensor measurements besides built-in temperature measurement capabilities sporting advanced compensation techniques.

Metamorphosing to data acquisition. Vendors have successfully transformed the basic multimeter into a data acquisition system, enabling some very challenging applications, such as ageing tests, burn-in tests, batch testing and many more. Rigol's M300 is used for applications ranging from refrigerator testing to steel mills.

— *Shrinath Kulkarni, managing director - South Asia and Middle East, Rigol Technologies Inc.*

Parameters to consider before you buy that multimeter

- **Conformity to safety standards.** Make sure the digital multimeter has CAT certification, CE marking, good IP rating, overload protection, surge protection and fire-resistant casing.
- **Should meet your measurement needs.** The application where your digital multimeter will be used should be the cornerstone of your purchase. Buy one keeping current and potential future requirements in mind.
- **Auto-range multimeters make it easy.** Manual ranging is a very old technology. By choosing auto-range multimeters, accidentally selecting the wrong range can be avoided.
- **Accuracy of measurements.** If you are a professional who cannot compromise on accuracy, then a multimeter from a reputed vendor is a must. For a hobbyist tinkering with circuits, where accuracy is not of great significance, low-cost multimeters would suffice, although most of these do not conform to safety standards.
- **Dimension, weight and long battery life.** Crucial parameters for those who like to carry their multimeter along everywhere, like field testing, for instance.
- **Miscellaneous.** Range of measurement, number of digits in the display (3½, 4, 4½, etc), safety ratings, rugged body, remote connectivity, well-lit display, warranty, service and affordable spare parts are some other factors that could be considered as well.

to the IR port, and U1115A remote-logging display can be used to view and log test measurements with an extended range of up to 100 metres. Yokogawa's TY series of multimeters have data logging capabilities that enable the user with hands-free data storage and provides online graphical plots as well.

Flir's DM93, too, features wireless functionality connecting to two sets of devices—FLIR thermal cameras and mobile platforms—all via Bluetooth. Sam Ruback, product manager - test and measurement, FLIR, says, "Connecting to your iOS, Android and even your PC with FLIR software tools adds convenience and safety for your problematic testing targets." He adds, "We have an exclusive Meterlink technology that integrates DM93's digital multimeter

Notable Multimeter Brands

International	Indian
BK Precision	HTC Instruments
Flir	Kusam-MECO
Fluke	MetroQ
GW Instek	Motwane
Keithley Instruments	Rishabh
Keysight	
Rigol	
Tektronix	
Yokogawa	

measurement data onto FLIR Meterlink-enabled cameras for advanced analysis."

Touch-screen interface. Slowly, vendors are starting to incorporate a touch-screen user interface to their T&M equipment. Be it oscilloscopes or digital multimeters, touch-screen is slowly trickling down to T&M tools with the aim of making these